

Graph-Based vs Transition-Based Dependency Parsing for Uzbek

A Cross-Architecture Study under Low-Resource Conditions

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- ▶ **Agglutinative**: one word stacks suffixes for case, tense, person, evidentiality.
- ▶ **Flexible SOV** word order with frequent center-embedding; null subjects; no articles.
- ▶ **Low-resource**: **1,184** total UD sentences — orders of magnitude less than English or Turkish.

Dependency parsing predicts a syntactic **head** and **relation** for every word — backbone of IE, MT, and sentiment analysis.

The suffix *is* the relation

bola + lar + ning

ROOT + PLUR + **GEN**

Miss the final suffix $\Rightarrow$ miss the parse.

1. **First cross-architecture comparison** of a **graph-based** (Stanza, DeepBiaffine) and a **transition-based** (spaCy, arc-eager) parser for Uzbek — both using the same **TahrirchiBERT** encoder.
2. **Architecture-dependent effects of cross-treebank augmentation**: merging two genre-complementary UD treebanks yields dramatically different LAS gains per parser type.
3. **Per-relation and per-feature error analyses** revealing graph-based parsing excels at structurally complex relations; transition-based parsing tags better locally.

Models & code: huggingface.co/Sanatbek/uzudt
github.com/SanatbekMatlatipov/robust-parsing-uzbek

UD_Uzbek-UzUDT — 684 sents ($\sim 7.8k$ tokens), fiction & academic; six annotators on INCEpTION, IAA $> 95\%$ (UPOS/lemma).

UD_Uzbek-UT — 500 sents ($\sim 5.9k$ tokens), news & fiction; semi-automatic + manual correction.

Both re-split to UD proportions ($\sim 66/7/27\%$).

Split	UzUDT	UT	Merged
Train	451	330	781
Dev	45	33	78
Test	188	137	325
Total	684	500	1,184

Table: Data splits.

Feature	UzUDT	UT	Benefit
Genres	Fic, Acad	News, Fic	3 genres
PROPN (%)	0.3	5.2	NER coverage
advcl count	444	106	Clause struct.
compound:lvc	21	215	LVC patterns
Morph. features	55	47	Feature union

Table: Key distributional differences.

Merging is not just *more* data — it is **broader** data: UT brings proper nouns & light-verb compounds; UzUDT brings adverbial clauses, possessor agreement, and evidentiality.

Stanza (graph-based)

- ▶ Two-stage: BiLSTM tagger → DeepBiaffine parser.
- ▶ Scores *all* head-dependent pairs (biaffine).
- ▶ BERT fused by **last-subword** selection.

Aspect	Stanza	spaCy
Parser	Graph	Transition
Strategy	Biaffine	Arc-eager
Training	Two-stage	Joint
Fusion	Last-sub	Mean
Max steps	50k	20k

spaCy (transition-based)

- ▶ Joint end-to-end; arc-eager transitions.
- ▶ Custom `spacy.blank("uz")` language class.
- ▶ BERT fused by `reduce_mean` pooling.

Table: Architectural comparison.

Surface	WordPiece split	Morphemes
<i>bolalarning</i>	bola ##lar ##ning	ROOT+PLUR+ GEN
<i>eshigini</i>	esh ##igi ##ni	ROOT+POSS.3+ ACC
<i>ko'rsatib</i>	ko'r ##sat ##ib	ROOT+CAUS+ CONV

Table: WordPiece decomposition: the last subword (bold) carries the UD-annotated morpheme.

Mean pooling averages all pieces and *dilutes* the suffix signal. **Last-subword** preserves Case/Tense/converb cues — the features that drive nsubj, obj, obl, xcomp attachment. Ablation: **+2.95 UAS / +2.64 LAS** (E2.1 vs E3.1), with UPOS essentially unchanged.

Stanza: E1.1/E1.2 (FastText), E2.1/E2.2 (BERT + last-sub), E3.1 (BERT + mean).

spaCy: S1.1/S1.2 (BERT).

.1 = UzUDT only .2 = Merged.

Default hyper-parameters for a fair
“practitioner” comparison; single NVIDIA RTX
A6000 (CUDA 12.4).

Metrics (UD CoNLL 2018)

UPOS, XPOS, UFeats (morphology),
UAS (unlabeled), LAS (labeled).

Stanza trains the parser head on *frozen*
BERT; spaCy *fine-tunes* the full
transformer jointly.

Run	Embed.	Data	UPOS	XPOS	UFeat	UAS	LAS
E1.1	FastText	UzUDT	79.19	79.81	66.61	69.57	51.24
E1.2	FastText	Merg.	80.26	83.20	66.98	72.27	62.40
E2.1	BERT _l	UzUDT	82.45	80.90	65.37	72.05	54.19
E2.2	BERT_l	Merg.	85.08	84.72	71.09	72.39	63.81
E3.1	BERT _m	UzUDT	82.76	81.37	65.22	69.10	51.55

Table: Stanza (graph-based).

Run	UPOS	UAS	LAS
S1.1	86.50	67.72	45.35
S1.2	89.18	66.81	47.11

Table: spaCy (transition).

BERT over FastText: **+3.3**–**+4.8** UPOS.
 Merging: single largest gain
 (**+9.6**–**+11.2** LAS).

Metric	Stanza	spaCy	Δ
UPOS	85.08	89.18	+4.10
XPOS	84.72	88.24	+3.52
UFeats	71.09	65.48	-5.61
UAS	72.39	66.81	-5.58
LAS	63.81	47.11	-16.70

Architecture-dependent trade-off

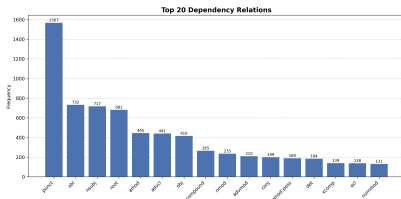
- ▶ spaCy tags better (joint training).
- ▶ Stanza parses **far** better: **+16.70 LAS** — among the largest gaps.
- ▶ Higher LAS *despite* lower UPOS: architecture, not tagging, is key.

Table: Cross-architecture comparison.

Metric	Stanza	spaCy
UPOS	+2.63	+2.68
XPOS	+3.82	+1.52
UFeats	+5.72	+14.93
UAS	+0.34	-0.91
LAS	+9.62	+1.76

Table: Effect of cross-treebank data.

- ▶ POS gain is **architecture-agnostic** ($\sim +2.6$ UPOS both).
- ▶ Stanza's graph parser exploits structural diversity: **+9.62 LAS** vs only +1.76 for spaCy.
- ▶ spaCy's joint objective amplifies morphology: **+14.93 UFeats** — gradients from tagger *and* parser simultaneously.



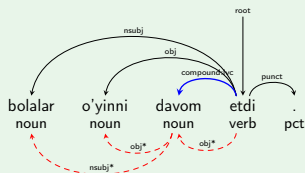
Top-20 relation frequencies (combined corpus).

Relation	P	R	F1
nummod	75.0	86.5	80.4
root	72.9	74.5	73.7
case	84.4	58.1	68.8
obj	58.6	65.1	61.7
nsubj	47.0	47.3	47.2
obl	47.6	46.7	47.1
compound:lvc	36.7	54.6	43.9
nmod	27.0	34.2	30.2
conj	25.8	14.9	18.9

Table: spaCy S1.2 per-relation LAS F1 (%).

Success — UAS = LAS = 100%

Failure — compound:lvc



Gold (blue) vs predicted errors (red dashed).

- ▶ **Pronounced trade-off:** transition-based tags better (+4.10 UPOS); graph-based parses far better (**+16.70 LAS**).
- ▶ **Cross-treebank augmentation** is the single most impactful lever — graph-based parsers exploit it best (**+9.62** vs +1.76 LAS).
- ▶ **Recommendation:** graph-based parser + monolingual BERT + last-subword fusion when parsing accuracy is the goal.

Future work: multi-seed testing; broader frameworks (UDPipe 2.0, Trankit, MaChAmp) and encoders (UzbBERT, XLM-R); LLM-based augmentation; parser ensembling.

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Questions welcome.